

MYSQL Queries

Project name: **Bank Loan Approval & Risk Analysis Dashboard**

Queries and their Result:

1.SELECT * FROM project1.loan_approval_dataset;

Result:

loan_id	no_of_dependents	education	self_employed	income_annum	loan_amount	loan_term	cibil_score
1	2	Graduate	No	9600000	29900000	12	778
2	0	Not Graduate	Yes	4100000	12200000	8	417
3	3	Graduate	No	9100000	29700000	20	506
4	3	Graduate	No	8200000	30700000	8	467
5	5	Not Graduate	Yes	9800000	24200000	20	382
6	0	Graduate	Yes	4800000	13500000	10	319
7	5	Graduate	No	8700000	33000000	4	678

residential_assets_value	commercial_assets_value	luxury_assets_value	bank_asset_value	loan_status	Total Assets
2400000	17600000	22700000	8000000	Approved	50700000
2700000	2200000	8800000	3300000	Rejected	17000000
7100000	4500000	33300000	12800000	Rejected	57700000
18200000	3300000	23300000	7900000	Rejected	52700000
12400000	8200000	29400000	5000000	Rejected	55000000
6800000	8300000	13700000	5100000	Rejected	33900000
22500000	14800000	29200000	4300000	Approved	70800000

#Total Loan Applications

2.SELECT COUNT(*) AS total_applications

FROM project1.loan_approval_dataset;

Result:

total_applications
4269

#Loan Approval vs Rejection Count

3.SELECT loan_status,

COUNT(*) AS total_loans

FROM project1.loan_approval_dataset

GROUP BY loan_status;

Result:

loan_status	total_loans
Approved	2656
Rejected	1613

#Loan Approval Rate

```
4.SELECT  
  
ROUND(  
SUM(CASE WHEN loan_status = 'Approved' THEN 1 ELSE 0 END)  
/ COUNT(*) * 100,2  
) AS approval_rate_percentage  
FROM project1.loan_approval_dataset;
```

Result:

approval_rate_percentage
62.22

#Average Income by Loan Status

```
5.SELECT loan_status,  
  
ROUND(AVG(income_annum),2) AS avg_income  
FROM project1.loan_approval_dataset  
GROUP BY loan_status;
```

Result:

loan_status	avg_income
Approved	5025903.61
Rejected	5113825.17

#Loan Applications by Education

```
6.SELECT education,  
  
COUNT(*) AS total_applications  
FROM project1.loan_approval_dataset  
GROUP BY education  
ORDER BY total_applications DESC;
```

Result:

education	total_applications
Graduate	2144
Not Graduate	2125

#Average Credit Score by Loan Status

```
7.SELECT loan_status,  
  
ROUND(AVG(cibil_score),2) AS avg_credit_score  
  
FROM project1.loan_approval_dataset  
  
GROUP BY loan_status;
```

Result:

loan_status	avg_credit_score
Approved	703.46
Rejected	429.47

#Average Loan Amount by Employment Type

```
8.SELECT self_employed,  
  
ROUND(AVG(loan_amount),2) AS avg_loan_amount  
  
FROM project1.loan_approval_dataset  
  
GROUP BY self_employed;
```

Result:

self_employed	avg_loan_amount
No	15120245.40
Yes	15146465.12

#Loan Amount by Education

```
9.SELECT education,  
  
ROUND(AVG(loan_amount),2) AS avg_loan  
  
FROM project1.loan_approval_dataset  
  
GROUP BY education;
```

Result:

education	avg_loan
Graduate	15229151.12
Not Graduate	15036894.12

#Risk Segmentation

10.SELECT

CASE

WHEN cibil_score >= 750 THEN 'Low Risk'

WHEN cibil_score BETWEEN 650 AND 749 THEN 'Medium Risk'

ELSE 'High Risk'

END AS risk_category,

COUNT(*) AS total_customers

FROM project1.loan_approval_dataset

GROUP BY risk_category;

Result:

risk_category	total_customers
Low Risk	1056
High Risk	2468
Medium Risk	745

#Total Assets Calculation

11.SELECT loan_id,

(residential_assets_value +

commercial_assets_value +

luxury_assets_value +

bank_asset_value) AS total_assets

FROM project1.loan_approval_dataset;

Result:

loan_id	total_assets
1	50700000
2	17000000
3	57700000
4	52700000
5	55000000
6	33900000
7	70800000
8	36700000

#Assets vs Loan Approval

```
12.SELECT loan_status,  
AVG(residential_assets_value +  
commercial_assets_value +  
luxury_assets_value +  
bank_asset_value) AS avg_total_assets  
FROM project1.loan_approval_dataset  
GROUP BY loan_status;
```

Result:

loan_status	avg_total_assets
Approved	32377296.6867
Rejected	32831122.1327